FOR A DOMINANT RING MAP THERE IS A SUBALGEBRA WHOSE SPECTRUM MAP IS A UNIVERSAL HOMEOMORPHISM AND WHICH CONTAINS THE IMAGES OF ALL SUCH FACTORIZATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-relative-weak-normalization-algebra is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 14747-14758 and proof span 14760-14810 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. Tag self-check per LANE_BRIEF 10.5 (5): the label lemma-relative-weak-normalization-algebra resolves in the project's own tag list to canonical tag 0H3J (line 21319 of the index, full label morphisms-lemma-relative-weak-normalization-algebra). reference/public-items.jsonl holds no row whose primary source_file is morphisms.tex, so the intersection with the same-chapter public tag ids is empty and the candidate passes.

Editorial note. The gate was run as: python operations/agent-authoring/tools/precard_gate.py –source morphisms.tex –label lemma-relative-weak-normalization-algebra –start 14747 –end 14758 –proof-start 14760 –proof-end 14810 –local-id 34093, giving RESULT=OK with all four span guards passing, no public interval row for this chapter and occupancy FREE for morphisms.tex 14747 14810. Spans were recomputed live from the label in the pinned chapter file and the located text matched the pool byte counts. The proof span contains a commutative diagram drawn with xymatrix, which the retained 88-line chapter preamble supports.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- Reduction of universal homeomorphism statements to the associated reduced rings, cited from the same chapter. Cited at proof line 14761. Lets the whole argument be carried out after replacing both rings by their reductions.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- For a reduced ring map inducing a dominant map of spectra the base is contained in the target, cited from the algebra chapter. Cited at proof lines 14779-14780. Makes the reduced case an inclusion of rings, which is what the structure result for universal homeomorphisms requires as input.
- The structure of a finite type algebra inducing a universal homeomorphism: every element lies in an extension generated by finitely many elements satisfying square or cube relations or p-th power relations over the preceding ones, cited from the same chapter. Cited at proof lines 14783-14790. Shows that the image of any admissible factorization is contained in the subalgebra constructed below, which is property (2) of the statement.

Source reference. Original lemma 00FL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 575–575. Same-chapter reference used in the retained author text at source line(s) 1301, 11842, 11946, 12203, 14779. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 054M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6645–6645. Same-chapter reference used in the retained author text at source line(s) 11413, 11943, 14761. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original proposition 0CNE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15889–15889. Same-chapter reference used in the retained author text at source line(s) 11950, 14781, 14808, 14874. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $A \to B$ be a ring map inducing a dominant morphism* $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ *of spectra. There exists an A-subalgebra $B' \subset B$ such that*

(1) $\mathrm{Spec}(B') \to \mathrm{Spec}(A)$ *is a universal homeomorphism,*
(2) *given a factorization $A \to C \to B$ such that* $\mathrm{Spec}(C) \to \mathrm{Spec}(A)$ *is a universal homeomorphism, the image of $C \to B$ is contained in B'.*

Proof. We will use Lemma 054M without further mention. Consider the commutative diagram

$$\begin{array}{ccc} B & \longrightarrow & B_{red} \\ \uparrow & & \uparrow \\ A & \longrightarrow & A_{red} \end{array}$$

For any factorization $A \to C \to B$ of $A \to B$ as in (2), we see that $A_{red} \to C_{red} \to B_{red}$ is a factorization of $A_{red} \to B_{red}$ as in (2). It follows that if the lemma holds for $A_{red} \to B_{red}$ and produces the A_{red}-subalgebra $B'_{red} \subset B_{red}$, then setting $B' \subset B$ equal to the inverse image of B'_{red} solves the lemma for $A \to B$. This reduces us to the case discussed in the next paragraph.

Assume A and B are reduced. In this case $A \subset B$ by Algebra, Lemma 00FL. Let $A \to C \to B$ be a factorization as in (2). Then we may apply Proposition 0CNE to $A \subset C$ to see that every element of C is contained in an extension $A[c_1, \ldots, c_n] \subset C$ such that for $i = 1, \ldots, n$ we have

(1) $c_i^2, c_i^3 \in A[c_1, \ldots, c_{i-1}]$, or

(2) there exists a prime number p with $pc_i, c_i^p \in A[c_1, \ldots, c_{i-1}]$.

Thus property (2) holds if we define $B' \subset B$ to be the subset of elements $b \in B$ which are contained in an extension $A[b_1, \ldots, b_n] \subset B$ such that (*) holds: for $i = 1, \ldots, n$ we have

(1) $b_i^2, b_i^3 \in A[b_1, \ldots, b_{i-1}]$, or

(2) there exists a prime number p with $pb_i, b_i^p \in A[b_1, \ldots, b_{i-1}]$.

There are only two things to check: (a) B' is an A-subalgebra, and (b) $\mathrm{Spec}(B') \to \mathrm{Spec}(A)$ is a universal homeomorphism. Part (a) follows because given $n \geq 0$ and $b_1, \ldots, b_n \in B$ satisfying (*) and $m \geq 0$ and $b'_1, \ldots, b'_m \in B$ satisfying (*), the integer $n + m$ and $b_1, \ldots, b_n, b'_1, \ldots, b'_m \in B$ also satisfies (*). Finally, part (b) holds by Proposition 0CNE and our construction of B'. □